

EE 316

Section 3

Pre Lab 05

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Basic Filters and Frequency Response

Introduction:

The goal of this laboratory is to study the frequency characteristics of low-pass and high-pass filters. The effects of input frequency on signal amplitude and phase angle are examined through theoretical calculations and experimental measurements. The theoretical gain and phase angle are calculated for a range of frequencies, and the cutoff frequency is determined and used to identify the point where the filter gain is approximately -3 dB.

Theoretical Analysis:

Equations 1 and 2, given in the lab manual, were used to find the gain and phase angle for a low pass filter with $V_{in} = 4V$, $R = 1k\Omega$, and $C = 1\mu F$. Equation 3 was used to find the cutoff frequency.

$$\left| \frac{V_{out}}{V_{in}} \right| = \frac{1}{\sqrt{1 + (2\pi fRC)^2}} \Rightarrow G = 20 \log \left(\frac{1}{\sqrt{1 + (2\pi fRC)^2}} \right) dB \quad (1)$$

$$\theta = \tan^{-1}(-2\pi fRC) \quad (2)$$

$$f_c = \frac{1}{2\pi RC} \quad (3)$$

Equations 4 and 5 show equations 1 and 2 evaluated at $f = 25Hz$. These calculations were then repeated for varying values of f and used to complete Table 1. Equation 6 shows the cutoff frequency calculation. Figures 1 and 2 show the plots of gain and phase angle versus frequency.

$$G = 20 \log \left(\frac{1}{\sqrt{1 + (2\pi * 25 * 1000 * 10^{-6})^2}} \right) dB = -0.1059 dB \quad (4)$$

$$\theta = \tan^{-1}(-2\pi * 25 * 1000 * 10^{-6}) = -8.9271^\circ \quad (5)$$

$$f_c = \frac{1}{2\pi * 1000 * 10^{-6}} = 159.1594 Hz \quad (6)$$

f (Hz)	Theoretical	
	Gain (dB)	Phase Angles (degrees)
25	-0.1059	-8.9271
50	-0.4088	-17.4406
75	-0.8709	-25.2316
100	-1.4451	-32.1419
150	-2.7606	-43.3038
200	-4.1147	-51.4881
300	-6.5830	-62.0533
500	-10.3621	-72.3432
600	-11.8219	-75.1439
700	-13.0845	-77.1908
800	-14.1940	-78.7483
900	-15.1822	-79.9716
1000	-16.0722	-80.9570

Table #1: Low Pass Filter Theoretical Results

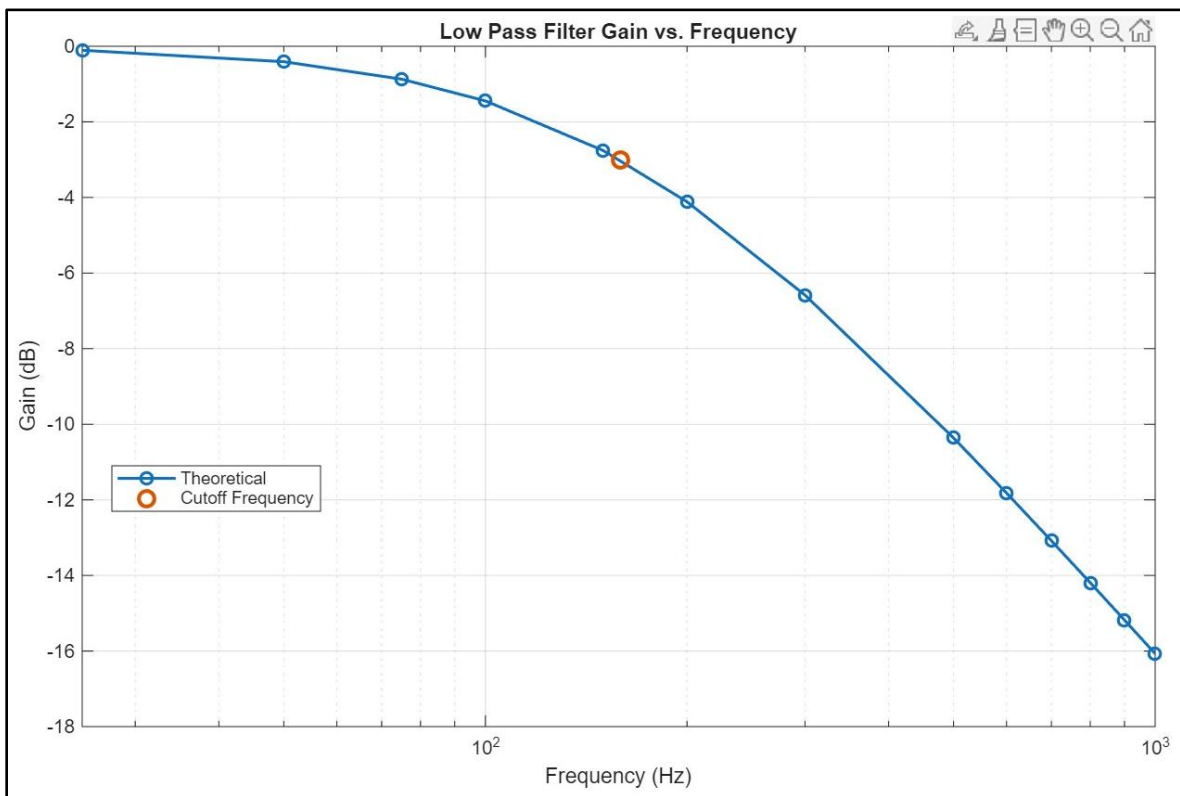


Figure #1: Gain vs. Frequency (LPF)

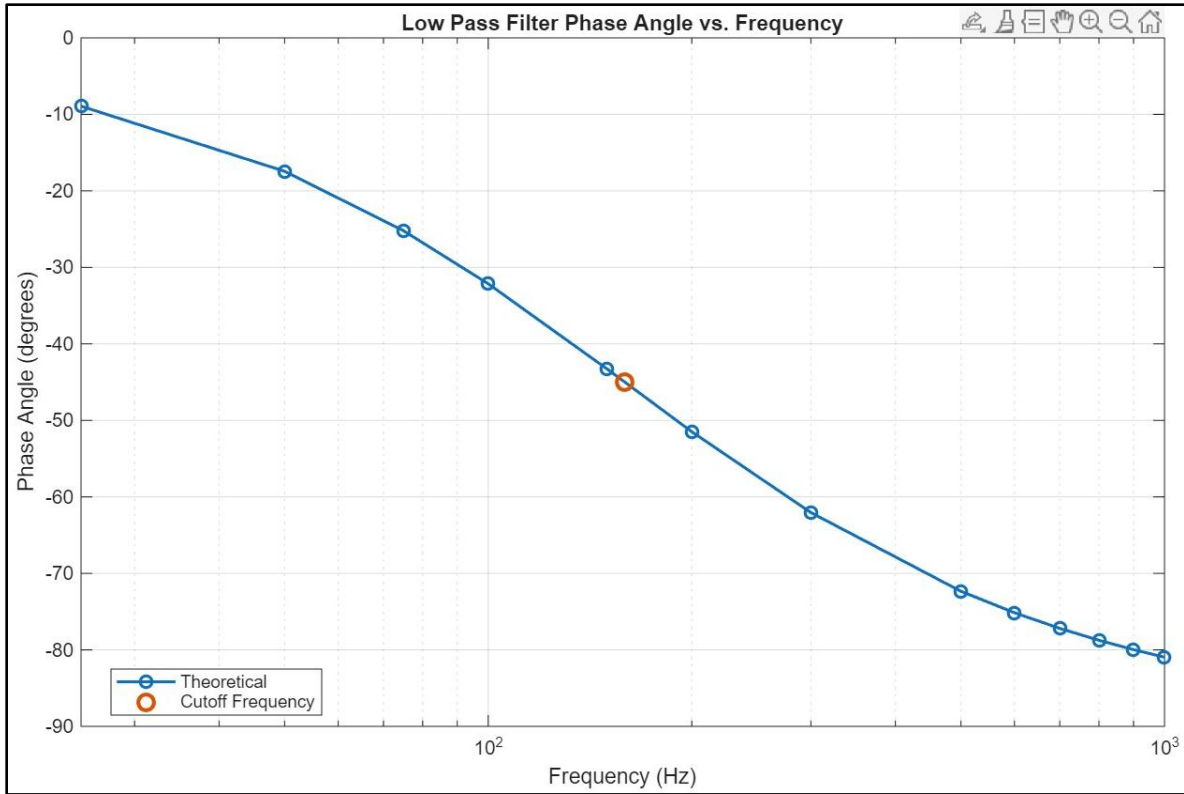


Figure #2: Phase Angle vs. Frequency (LPF)

Equations 7 and 8, given in the lab manual, were used to find the gain and phase angle for a high pass filter with $V_{in} = 4V$, $R = 1k\Omega$, and $C = 1\mu F$. Equation 9 was used to find the cutoff frequency. It is important to note that Equation 9 matches Equation 3, which was used to find the cutoff frequency of the low pass filter.

$$\left| \frac{V_{out}}{V_{in}} \right| = \frac{2\pi fRC}{\sqrt{1 + (2\pi fRC)^2}} \Rightarrow G = 20 \log \left(\frac{2\pi fRC}{\sqrt{1 + (2\pi fRC)^2}} \right) dB \quad (7)$$

$$\theta = \tan^{-1} \left(\frac{1}{2\pi fRC} \right) \quad (8)$$

$$f_c = \frac{1}{2\pi RC} \quad (9)$$

Equations 10 and 11 show equations 7 and 8 evaluated at $f = 25Hz$. These calculations were then repeated for varying values of f and used to complete Table 2. Equation 12 shows the cutoff frequency calculation. Figures 3 and 4 show the plots of gain and phase angle versus frequency.

$$G = 20 \log \left(\frac{2\pi * 25 * 1000 * 10^{-6}}{\sqrt{1 + (2\pi * 25 * 1000 * 10^{-6})^2}} \right) dB = -16.1835 \quad (10)$$

$$\theta = \tan^{-1} \left(\frac{1}{2\pi * 25 * 1000 * 10^{-6}} \right) = 81.0730 \quad (11)$$

$$f_c = \frac{1}{2\pi * 1000 * 10^{-6}} = 159.1594 \text{ Hz} \quad (6)$$

f (Hz)	Theoretical	
	Gain (dB)	Phase Angles (degrees)
25	-16.1835	81.0730
50	-10.4658	72.5594
75	-7.4061	64.7684
100	-5.4815	57.8581
150	-3.2752	46.6962
200	-2.1305	38.5119
300	-1.0770	27.9467
500	-0.4191	17.6568
600	-0.2953	14.8561
700	-0.2189	12.8092
800	-0.1686	11.2517
900	-0.1337	10.0284
1000	-0.1086	9.0431

Table #2: High Pass Filter Theoretical Results

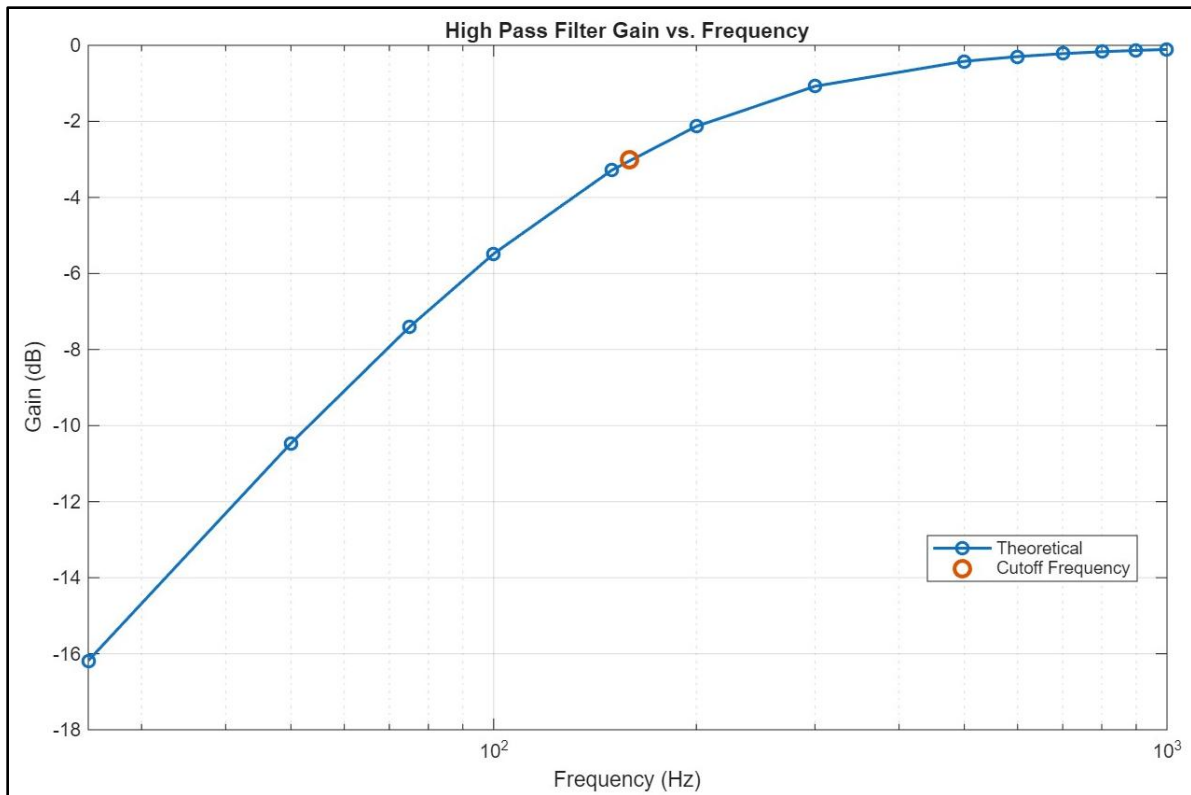


Figure #3: Gain vs. Frequency (HPF)

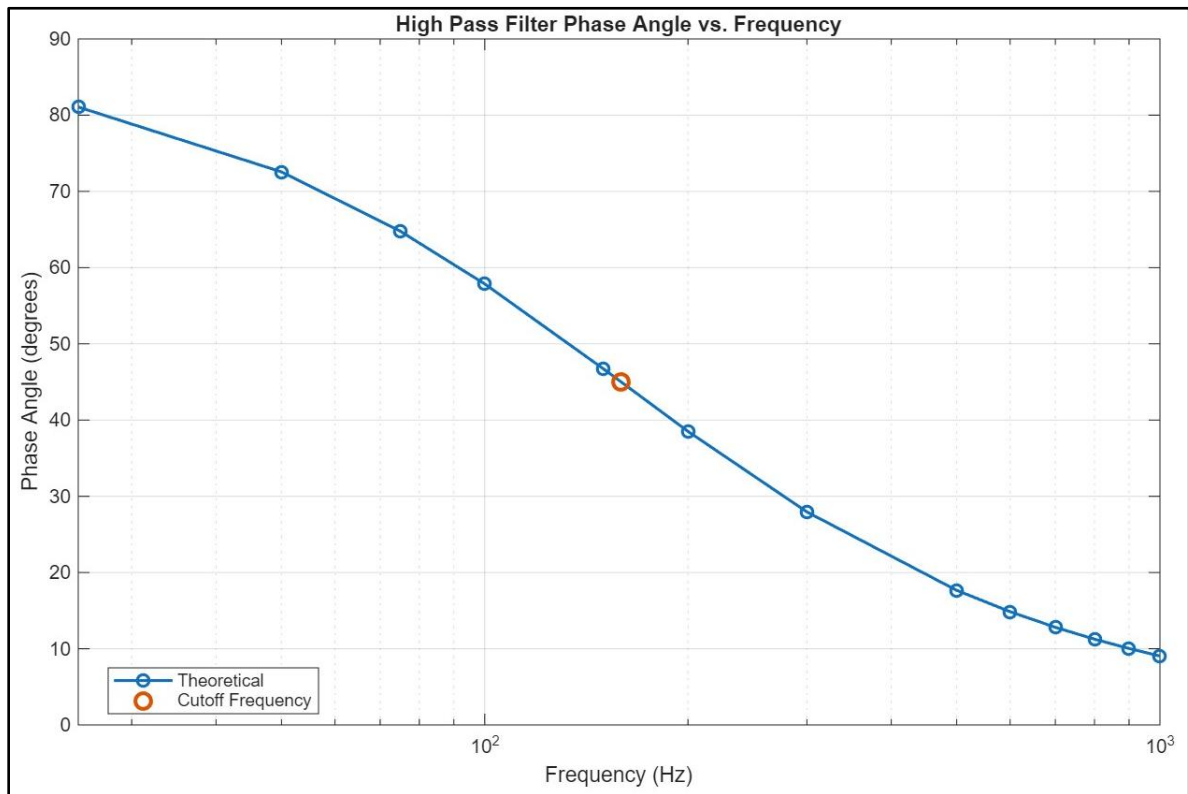


Figure #4: Phase Angle vs. Frequency (HPF)

Simulations and Results:

LTSpice was used to simulate both a low pass and high pass filter with $V_{in} = 4V_{pk-pk}$, $R = 1k\Omega$, and $C = 1\mu F$. Figures 5 and 6 show the schematics given in the lab manual for the LPF and HPF. Figure 7 shows the LTSpice schematic used to simulate both filters simultaneously. Figure 8 shows the Bode plot for both simulated filters with the simulated -3dB point marked. Tables 3 and 4 show the simulation results for the LPF and HPF respectively.

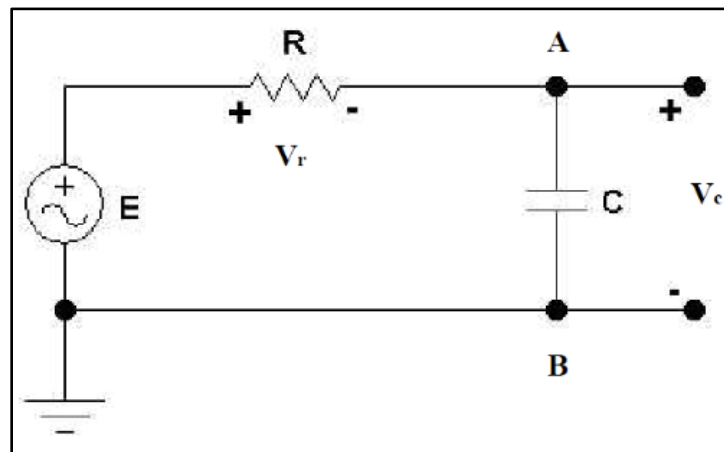


Figure #5: Low Pass Filter Schematic (Lab Manual)

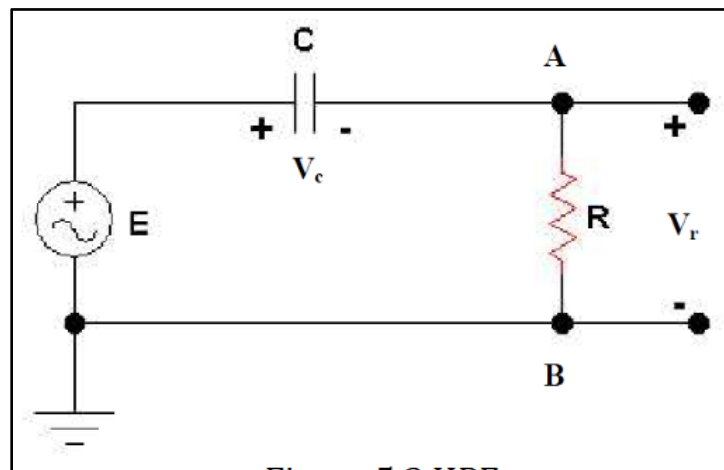


Figure #6: Low Pass Filter Schematic (Lab Manual)

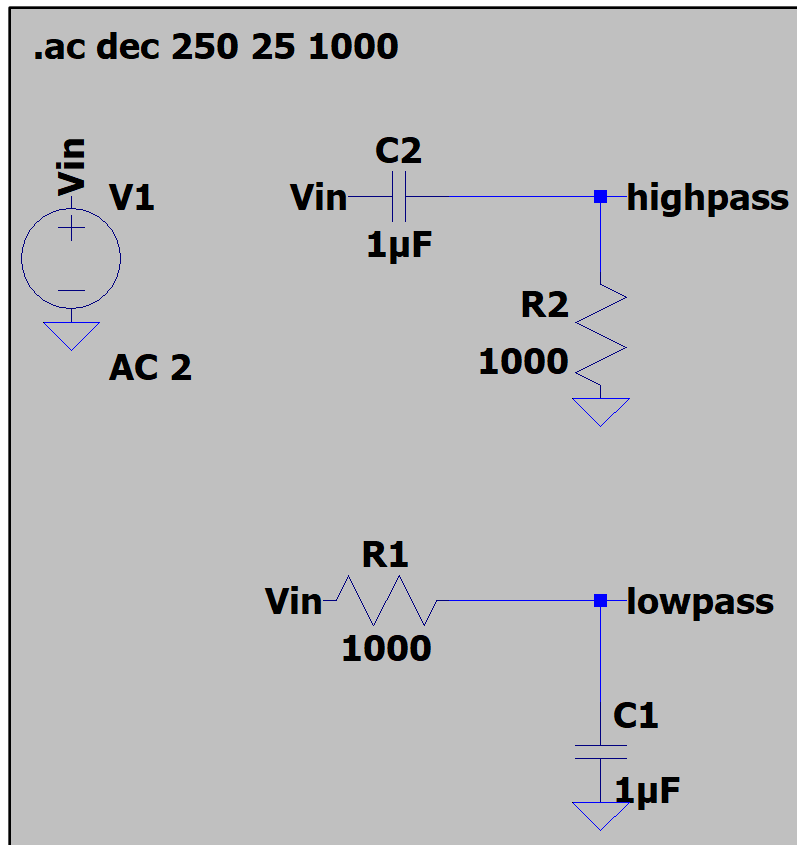


Figure #7: LTSpice Schematic (LPF and HPF)

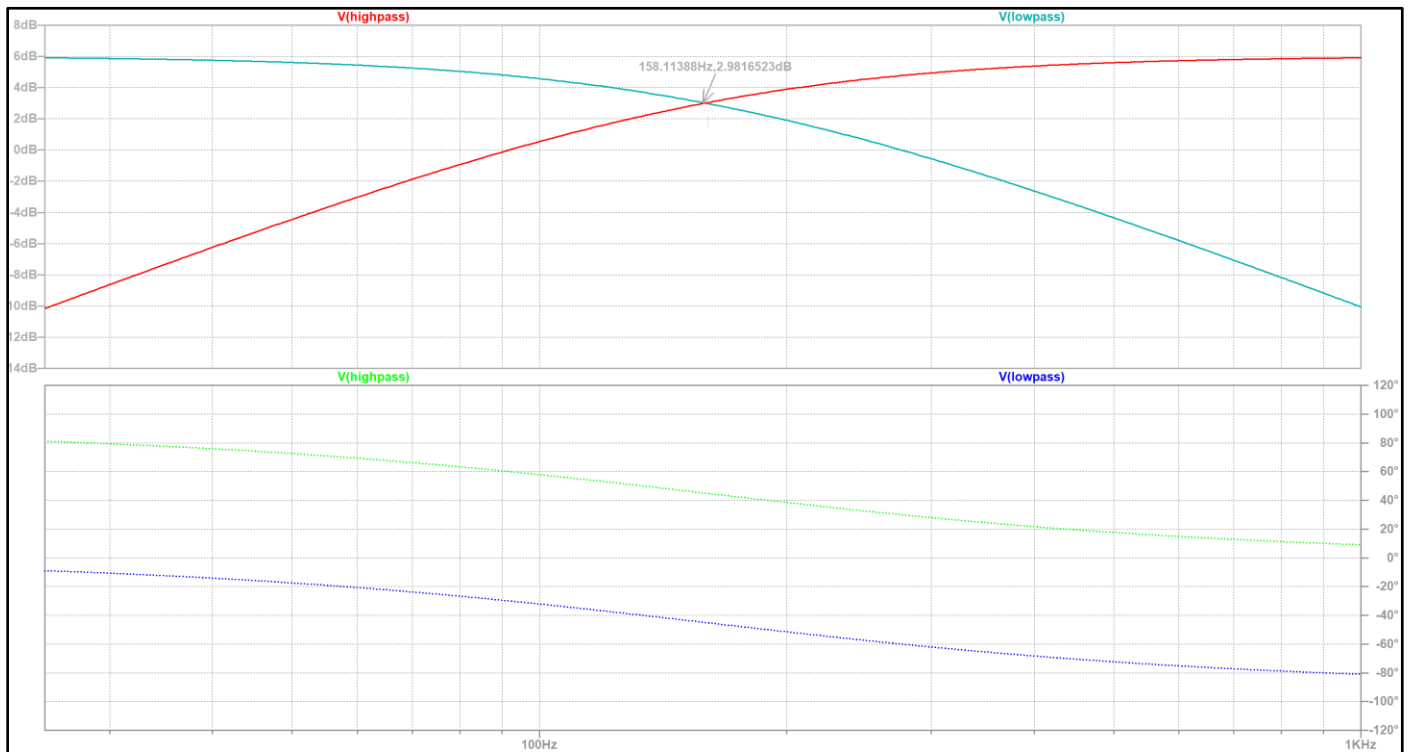


Figure #8: LTSpice Bode Plot (LPF and HPF)

	Simulation	
f (Hz)	Gain (dB)	Phase Angles (degrees)
25	-0.10586154	-8.9271434
50	-0.4087871	-17.440703
75	-0.8709666	-25.231744
100	-1.4450701	-32.141908
150	-2.7606321	-43.303809
200	-4.114777	-51.488062
300	-6.5830569	-62.053199
500	-10.362149	-72.343103
600	-11.82194	-75.143819
700	-13.084461	-77.190656
800	-14.193975	-78.748199
900	-15.182185	-79.971464
1000	-16.072235	-80.956939

Table #3: Low Pass Filter Simulation Results

	Simulation	
f (Hz)	Gain (dB)	Phase Angles (degrees)
25	-16.183464	81.072857
50	-10.46579	72.559297
75	-7.406144	64.768256
100	-5.4814727	57.858092
150	-3.2752096	46.696191
200	-2.1305798	38.511938
300	-1.0770344	27.946801
500	-0.41915173	17.656897
600	-0.29531781	14.856181
700	-0.21890279	12.809344
800	-0.16857801	11.251801
900	-0.13373748	10.028536
1000	-0.1086379	9.0430611

Table #4: High Pass Filter Simulation Results